EXP 22

```
clc;
clear;
close all;

%% 1. System Parameters
N = 10000;              % Number of bits
SNR_dB = 10;            % Signal-to-noise ratio in dB

%% 2. Generate Random Binary Data
bits = randi([0 1], N, 1);

%% 3. BPSK Modulation
% Bit 0 -> -1
% Bit 1 -> +1
tx_symbols = 2*bits - 1;

%% 4. Add AWGN Noise
% For BPSK with unit bit energy, noise variance per
% real dimension is N0/2 = 1/(2*Eb/N0).
EbN0 = 10^(SNR_dB/10);
noise = sqrt(1/(2*EbN0)) * randn(N, 1);

rx_symbols = tx_symbols + noise;

%% 5. Constellation Diagram Before Noise
figure;
plot(real(tx_symbols), imag(tx_symbols), 'bo', ...
    'MarkerSize', 5);
grid on;
axis equal;
```

```
xlim([-1.5 1.5]);
ylim([-1.5 1.5]);
xlabel('In-Phase (I)');
ylabel('Quadrature (Q)');
title('BPSK Constellation Before Noise');

%% 6. Constellation Diagram After Noise
figure;
plot(real(rx_symbols), imag(rx_symbols), 'r.', ...
    'MarkerSize', 8);
grid on;
axis equal;
xlim([-2 2]);
ylim([-1.5 1.5]);
xlabel('In-Phase (I)');
ylabel('Quadrature (Q)');
title(['BPSK Constellation With AWGN, SNR = ', ...
    num2str(SNR_dB), ' dB']);

%% 7. BPSK Demodulation
detected_bits = rx_symbols > 0;

%% 8. Calculate Bit Error Rate
errors = sum(bits ~= detected_bits);
BER_simulated = errors/N;

% Theoretical BER for coherent BPSK in AWGN
BER_theoretical = 0.5 * erfc(sqrt(EbN0));

fprintf('BPSK Modulation Results\n');
fprintf('-----------------------\n');
```

```
fprintf('Number of bits      = %d\n', N);
fprintf('SNR (Eb/N0)         = %d dB\n', SNR_dB);
fprintf('Number of bit errors = %d\n', errors);
fprintf('Simulated BER       = %.6f\n', BER_simulated);
fprintf('Theoretical BER     = %.6e\n', BER_theoretical);

%% 9. BER Performance for Different SNR Values
SNR_range = 0:2:12;
BER_sim = zeros(size(SNR_range));
BER_theory = zeros(size(SNR_range));

for k = 1:length(SNR_range)
   ebn0 = 10^(SNR_range(k)/10);

   % Generate independent noise for each SNR
   n = sqrt(1/(2*ebn0)) * randn(N,1);
   received = tx_symbols + n;

   detected = received > 0;
   BER_sim(k) = sum(bits ~= detected)/N;
   BER_theory(k) = 0.5*erfc(sqrt(ebn0));
end

figure;
semilogy(SNR_range, max(BER_sim, 1/N), 'bo-', ...
   'LineWidth', 1.5, 'MarkerSize', 6);
hold on;
semilogy(SNR_range, BER_theory, 'r*-', ...
   'LineWidth', 1.5, 'MarkerSize', 6);
grid on;
xlabel('E_b/N_0 (dB)');
```

```
ylabel('Bit Error Rate (BER)');
title('BPSK BER Performance in AWGN');
legend('Simulated BER', 'Theoretical BER', ...
    'Location', 'southwest');
```

